

Machine Learning Practice Problems

Step-by-Step Complete Solutions & Explanations

Problem 1: Gradient Descent (1D)

Given the function: $f(x) = (x+4)^2$. Starting at $x_0 = 3$ with a learning rate $\eta = 0.01$.

(a) What is the gradient at $x_0 = 3$?

The gradient of a single-variable function is its derivative. Using the chain rule: $f'(x) = 2(x+4)$. Plugging in $x_0 = 3$ gives $f'(3) = 2(3+4) = 14$.

Answer: 14.

(b) What is the updated value of x after one step?

The update rule is $x_{\text{new}} = x_{\text{old}} - \eta \cdot f'(x_{\text{old}})$. Plugging in the values: $x_1 = 3 - (0.01 \cdot 14) = 3 - 0.14 = 2.86$.

Answer: 2.86.

Problem 2: Gradient Descent Variants

You have a dataset with $N = 1000$ samples.

(a) In batch gradient descent, how many samples are used to compute the gradient at each iteration?

Batch gradient descent looks at the entire dataset to calculate the gradient for a single step.

Answer: 1000 samples.

(b) In mini-batch gradient descent with a batch size of 50, how many samples are used per iteration?

Mini-batch looks at a small, randomly selected subset (batch) of the data for each step.

Answer: 50 samples.

(c) If you perform 10 iterations of mini-batch gradient descent (with replacement allowed), how many total sample usages occur?

Every iteration uses 50 samples. Across 10 iterations: $10 \times 50 = 500$.

Answer: 500 total sample usages.

Problem 3: Logistic Regression Calculations

Given Hypothesis $h_{\theta}(x) = 1 / (1 + e^{-(\theta^T x)})$, Parameter vector $\theta = [0.5, -0.25]^T$, Input vector $x = [1, 2]^T$, and True label $y = 1$.

(a) Compute $h_{\theta}(x)$.

First, compute the dot product: $\theta^T x = (0.5 \times 1) + (-0.25 \times 2) = 0.5 - 0.5 = 0$. Plug this into the sigmoid function: $h_{\theta}(x) = 1 / (1 + e^0) = 1 / (1 + 1) = 0.5$.

Answer: 0.5.

(b) Calculate the logistic regression cost for this example.

The cost formula is: $\text{Cost} = -y \log(h_{\theta}(x)) - (1 - y) \log(1 - h_{\theta}(x))$. Since $y = 1$, $\text{Cost} = -1 \cdot \log(0.5) = -\ln(0.5) \approx 0.693$.

Answer: 0.693 (or $-\ln(0.5)$).

Problem 4: Gaussian Discriminant Analysis (GDA)

Given class-conditional densities $p(x|y=0) = N(1, 1)$ [$\mu_0=1, \sigma^2=1$] and $p(x|y=1) = N(3, 1)$ [$\mu_1=3, \sigma^2=1$], with priors $P(y=0) = 0.6$ and $P(y=1) = 0.4$. Observed feature value $x = 2$.

(a) Compute the likelihoods $p(x|y=0)$ and $p(x|y=1)$.

The PDF is $f(x) = (1 / \sqrt{2\pi\sigma^2}) * e^{-(x-\mu)^2 / 2\sigma^2}$. Since $\sigma^2 = 1$, it simplifies to $(1 / \sqrt{2\pi}) * e^{-(x-\mu)^2 / 2}$.

For $y=0$: $p(x=2|y=0) = (1/\sqrt{2\pi}) * e^{-(2-1)^2/2} = (1/\sqrt{2\pi}) * e^{-0.5} \approx 0.242$.

For $y=1$: $p(x=2|y=1) = (1/\sqrt{2\pi}) * e^{-(2-3)^2/2} = (1/\sqrt{2\pi}) * e^{-0.5} \approx 0.242$.

(b) Compute the posterior probability $P(y=1|x=2)$.

Using Bayes' Theorem: $P(y=1|x=2) = [p(x=2|y=1)P(y=1)] / [p(x=2|y=0)P(y=0) + p(x=2|y=1)P(y=1)]$. Since the likelihoods are equal, they cancel out: $P(y=1|x=2) = 0.4 / (0.6 + 0.4) = 0.4$.

Answer: 0.4.

Problem 5: Naive Bayes Classifier

Dataset: (Sunny, Hot) -> No, (Sunny, Hot) -> No, (Rainy, Mild) -> Yes, (Sunny, Mild) -> Yes, (Rainy, Hot) -> Yes. Totals: 5 samples (Yes=3, No=2).

(a) Explain what Naive Bayes assumes, and discuss one limitation and one advantage.

Assumption: Features are completely independent given the class label.

Advantage: Extremely fast, needs minimal data.

Limitation: The independence assumption is often violated in reality (e.g., sunny weather correlates with hot temperatures).

(b) Calculate the posterior probabilities for Weather = Sunny, Temperature = Mild. Which class is predicted?

Priors: $P(\text{Yes}) = 3/5$, $P(\text{No}) = 2/5$.

For Yes: $P(\text{Sunny}|\text{Yes}) = 1/3$, $P(\text{Mild}|\text{Yes}) = 2/3$. $\text{Score}(\text{Yes}) = 1/3 \times 2/3 \times 3/5 = 6/45 = 0.133$.

For No: $P(\text{Sunny}|\text{No}) = 2/2 = 1$, $P(\text{Mild}|\text{No}) = 0/2 = 0$. $\text{Score}(\text{No}) = 1 \times 0 \times 2/5 = 0$.

Answer: $P(\text{Yes}|\text{Sunny, Mild}) = 1.0$, $P(\text{No}|\text{Sunny, Mild}) = 0.0$. Predicted class is Yes.

Problem 6: Bagging vs. Boosting

(a) Explain the main conceptual difference focusing on training and combining.

Bagging trains base models in parallel on random bootstrap subsets of data and combines them via simple voting or averaging. Boosting trains base models sequentially, where each new model focuses on correcting errors from the previous one, combining them via weighted voting.

(b) Describe one advantage and one limitation of each method.

Bagging: Reduces variance (overfitting) but doesn't fix high bias.

Boosting: Reduces bias and achieves high accuracy, but is highly susceptible to overfitting on noisy data.

Problem 7: Multivariate Gradient Descent

Function: $f(\theta_1, \theta_2) = (\theta_1 - 2)^2 + 4(\theta_2 + 1)^2$. Starting point: $(4, 1)$, learning rate $\eta = 0.1$.

(a) Compute the gradient ∇f at the starting point.

Partial derivatives: $\partial f / \partial \theta_1 = 2(\theta_1 - 2)$, $\partial f / \partial \theta_2 = 8(\theta_2 + 1)$.

At $(4, 1)$: $\partial f / \partial \theta_1 = 2(4 - 2) = 4$; $\partial f / \partial \theta_2 = 8(1 + 1) = 16$.

Answer: $\nabla f(4,1) = [4, 16]^T$.

(b) Perform one step of gradient descent and report updated values.

$\theta_{\text{new}} = \theta_{\text{old}} - \eta \cdot \nabla f$

$\theta_1^{(1)} = 4 - (0.1 \times 4) = 3.6$

$\theta_2^{(1)} = 1 - (0.1 \times 16) = -0.6$.

Answer: $(3.6, -0.6)$.

(c) What is the global minimum of f ? Does it converge for any positive learning rate?

Global Minimum: Occurs when both squared terms are zero, at $(2, -1)$.

Convergence: No. If η is too large, the updates will overshoot the minimum and diverge to infinity.

Problem 8: Logistic Regression with Single Feature

Model: $h_{\theta}(x) = 1 / (1 + e^{-(\theta_0 + \theta_1 x)})$. Example: $(x,y) = (1,0)$, initial parameters $\theta_0 = 0$, $\theta_1 = 1$, $\eta = 0.5$.

(a) Compute the predicted probability $h_{\theta}(1)$.

Exponent: $-(0 + 1 \cdot 1) = -1$. $h_{\theta}(1) = 1 / (1 + e^{-1}) \approx 0.731$.

Answer: 0.731.

(b) Compute the binary cross-entropy loss for this example.

Since $y = 0$, Loss = $-\log(1 - h_{\theta}(x)) = -\log(1 - 0.731) = -\log(0.269) \approx 1.313$.

Answer: 1.313.

(c) Write the gradient descent update rule for θ_1 and compute the updated θ_1 .

Update rule: $\theta_1 \leftarrow \theta_1 - \eta \cdot (h_{\theta}(x) - y)x$.

Calculation: $\theta_1 \leftarrow 1 - 0.5 \cdot (0.731 - 0) \cdot 1 = 1 - 0.3655 = 0.6345$.

Answer: 0.6345.

Problem 9: Logistic Regression Decision Boundary

(a) Show that the decision boundary corresponds to the hyperplane $\theta^T x = 0$.

$$\text{Set } h_{\theta}(x) = 0.5 \Rightarrow 1 / (1 + e^{-(\theta^T x)}) = 0.5 \Rightarrow 1 + e^{-(\theta^T x)} = 2 \Rightarrow e^{-(\theta^T x)} = 1 \Rightarrow -\theta^T x = \ln(1) = 0 \Rightarrow \theta^T x = 0.$$

(b) Write the equation of the decision boundary for $\theta = [-3, 1, 2]^T$ with $x_0=1$ and describe it.

$$\text{Equation: } -3(1) + 1(x_1) + 2(x_2) = 0 \Rightarrow x_1 + 2x_2 = 3 \Rightarrow x_2 = -0.5x_1 + 1.5.$$

Description: A straight line in the x_1 - x_2 plane with a y-intercept at (0, 1.5) and a negative slope of -0.5.

(c) For a new point $x = [1, 0, 2]^T$, predict the class label (0 or 1).

$$\text{Compute } \theta^T x = (-3 \cdot 1) + (1 \cdot 0) + (2 \cdot 2) = -3 + 0 + 4 = 1.$$

Since $\theta^T x = 1 > 0$, the probability is > 0.5 .

Answer: Predict class 1.

Problem 10: Gaussian Discriminant Analysis (GDA) Math

Dataset: Class 0: $x \in \{1, 3\}$, Class 1: $x \in \{5, 7\}$.

(a) Write the maximum likelihood estimates for ϕ , μ_0 , and μ_1 .

$$\phi = n_1/n, \mu_0 = (\sum_{i:y=0} x^{(i)}) / n_0, \mu_1 = (\sum_{i:y=1} x^{(i)}) / n_1.$$

(b) Compute ϕ , μ_0 , μ_1 , and Σ for the given dataset.

$$n_0=2, n_1=2, n=4.$$

$$\phi = 2/4 = 0.5.$$

$$\mu_0 = (1+3)/2 = 2.$$

$$\mu_1 = (5+7)/2 = 6.$$

$$\Sigma \text{ (shared variance)} = [(1-2)^2 + (3-2)^2 + (5-6)^2 + (7-6)^2] / 4 = [1 + 1 + 1 + 1] / 4 = 1.$$

Answers: $\phi=0.5$, $\mu_0=2$, $\mu_1=6$, $\Sigma=1$.

(c) Classify a new point $x = 4$ using the GDA decision rule.

$x = 4$ lies exactly halfway between $\mu_0 = 2$ and $\mu_1 = 6$. Since class priors are identical (0.5) and the layout is perfectly symmetric, $x = 4$ lies exactly on the decision boundary.

Answer: Tie / Boundary point.

Problem 11: Naive Bayes Spam Classifier

Dataset: 6 total emails (3 Spam, 3 Not Spam). Spam: (Yes), (Yes), (No). Not Spam: (No), (No), (Yes).

(a) Estimate the class prior $P(\text{Spam})$ and the likelihoods.

$$P(\text{Spam}) = 3/6 = 0.5.$$

$$P(\text{"free"}=\text{Yes} \mid \text{Spam}) = 2/3.$$

$$P(\text{"free"}=\text{Yes} \mid \text{Not Spam}) = 1/3.$$

(b) A new email arrives with "free" = Yes. Predict the class.

$$\text{Score}(\text{Spam}) = 0.5 \times 2/3 = 1/3.$$

$$\text{Score}(\text{Not Spam}) = 0.5 \times 1/3 = 1/6.$$

Since $1/3 > 1/6$, predict Spam.

(c) Explain the zero-probability problem and how Laplace smoothing resolves it.

Problem: If a feature never appears with a class in the training data, its conditional probability becomes 0, which zeroes out the entire multiplication chain.

Solution: Laplace smoothing adds a small constant (usually +1) to the numerator and adjusts the denominator by the number of unique values, ensuring no probability is ever strictly zero.

Problem 12: Decision Tree Building

Dataset: 1. (Sunny, High) -> No; 2. (Sunny, Normal) -> Yes; 3. (Overcast, High) -> Yes; 4. (Overcast, Normal) -> Yes. Totals: 3 Yes, 1 No.

(a) Compute the entropy $H(\text{Play})$ of the full dataset.

$$H(\text{Play}) = -[(3/4)\log_2(3/4) + (1/4)\log_2(1/4)] \approx 0.811.$$

(b) Compute the information gain of splitting on Outlook and on Humidity.

Splitting on Outlook:

Sunny: 1 Yes, 1 No ($H=1.0$). Overcast: 2 Yes, 0 No ($H=0$). Remainder Entropy = 0.5. $IG(\text{Outlook}) = 0.811 - 0.5 = 0.311$.

Splitting on Humidity:

High: 1 Yes, 1 No ($H=1.0$). Normal: 2 Yes, 0 No ($H=0$). Remainder Entropy = 0.5. $IG(\text{Humidity}) = 0.811 - 0.5 = 0.311$.

(c) Which feature should be chosen as the root of the decision tree?

Both features yield the exact same Information Gain (0.311). Either can be chosen arbitrarily.

Answer: Either Outlook or Humidity.

Problem 13: Overfitting in Decision Trees

(a) Why might a fully grown tree perform poorly on unseen data? What phenomenon is this?

It memorizes the noise, random variations, and specific details of the training set rather than learning the general pattern. This is called Overfitting (High Variance).

(b) Describe two strategies (other than pruning) to mitigate this issue.

1. Setting a Maximum Depth: Restricts how deep the tree can branch.
2. Minimum Samples per Split: Requires a node to contain at least a certain number of samples to allow a split.

(c) Explain how Random Forest addresses overfitting.

It uses Bagging to train an ensemble of multiple trees on bootstrap data subsets and averages their votes to reduce variance, combined with Feature Randomization to ensure individual trees remain uncorrelated.

Problem 14: AdaBoost Step-by-Step

Given weighted error $\varepsilon_1 = 0.25$.

(a) Compute the learner weight α_1 .

$$\alpha_1 = 0.5 \cdot \ln((1 - 0.25) / 0.25) = 0.5 \cdot \ln(3) \approx 0.549.$$

Answer: 0.549.

(b) Write the general weight update rule.

Misclassified: $w_{\text{new}} = w_{\text{old}} \cdot e^{\alpha}$

Correctly Classified: $w_{\text{new}} = w_{\text{old}} \cdot e^{-\alpha}$

(c) Compute and normalize weights for 4 samples (initially 0.25; 1 & 2 misclassified, 3 & 4 correct).

Unnormalized:

Samples 1 & 2: $0.25 \times e^{(0.549)} \approx 0.433$ each.

Samples 3 & 4: $0.25 \times e^{(-0.549)} \approx 0.144$ each.

Sum = $0.433 + 0.433 + 0.144 + 0.144 = 1.154$.

Normalized:

Samples 1 & 2: $0.433 / 1.154 \approx 0.375$ each.

Samples 3 & 4: $0.144 / 1.154 \approx 0.125$ each.

Answer: Misclassified = 0.375, Correct = 0.125.

Problem 15: Non-Convex Optimization

Function: $f(\theta) = \theta^4 - 4\theta^2$.

(a) Find all stationary points and classify them.

$$f'(\theta) = 4\theta^3 - 8\theta = 0 \Rightarrow 4\theta(\theta^2 - 2) = 0 \Rightarrow \theta = 0, \theta = \sqrt{2}, \theta = -\sqrt{2}.$$

$$\text{Second derivative: } f''(\theta) = 12\theta^2 - 8.$$

At $\theta = 0$: $f''(0) = -8 (< 0) \rightarrow \text{Local Maximum}$.

At $\theta = \pm\sqrt{2}$: $f''(\pm\sqrt{2}) = 16 (> 0) \rightarrow \text{Local Minimums}$.

(b) Describe convergence behavior from $\theta_0 = 0.1$ and $\theta_0 = -0.1$.

From $\theta_0 = 0.1$, the gradient is negative, so it moves rightward and converges to the right valley at $\theta = \sqrt{2}$.

From $\theta_0 = -0.1$, it moves leftward and converges to the left valley at $\theta = -\sqrt{2}$.

Answer: Yes, they converge to different local minimums.

(c) Why do non-convex loss surfaces pose a challenge? Mention a strategy.

They contain multiple local minimums and saddle points where basic optimization can get stuck, failing to find the global minimum.

Strategy: Use Stochastic Gradient Descent (SGD) with Momentum or advanced optimizers like Adam.

Problem 16: Logistic Regression vs. GDA

(a) State one key assumption that GDA makes that logistic regression does not.

GDA assumes features are normally (Gaussian) distributed within each class, sharing an identical covariance matrix. It holds exactly when the true underlying feature distribution is Gaussian.

(b) Which classifier is expected to perform better with a small training set if assumptions hold?

GDA performs better because its structural assumptions act as a strong regularizer, allowing it to find the true decision boundary with less data.

(c) In practice, bag-of-words features are far from Gaussian. Does this favor logistic regression or GDA?

It heavily favors Logistic Regression. As a discriminative model, it makes no distribution assumptions and remains robust when features are discrete or non-Gaussian.